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Integrated disease study and symptom-based identification of vegetable crop pathogens in diverse agricultural landscapes

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Abstract- Vegetable crop diseases are a very important factor threatening the agricultural productivity and food security that need detailed field surveying for their management properly. Tomato, brinjal, cabbage and potato were selected for the study and a survey of four agricultural sites was conducted to determine major diseases. A total of eight diseases were recorded, amongst them tomatoes being highly diverse especially from Bamni Tola, Patamda. Fungi infections were dominant, showing necrotic spots and water soaked lesions. Solanaceae crops appeared especially vulnerable. Severity also ranged by location, suggesting intricate relationships. These results provide crucial baseline information for region-specific disease control, monitoring schemes and sustainable crop protection strategies in efforts to ensure regional food security.

Keywords: Vegetable diseases, disease identification, crop pathology, fungal diseases

INTRODUCTION

Vegetable crops are crucial on the world nutrition scale for supply of essential vitamins and minerals, dietary diversity and economic stability.[1] However, they perform very poorly due to diseases such as those caused by fungi, bacteria, viruses and nematodes. Pathogens of fungal origin are very common as a result of ideal environmental conditions such as humidity and moderate temperature.[2] Tomato, potato and brinjal crop diseases under the family Solanaceae are significantly feasible prospectus as they share their susceptibility risk.[3] Early Blight in tomato and Late Blight in potato are the primary cause of yield reduction.[4,5] Chemical Management is however non-sustainable,[6] thus necessitating integrated control. Community-situated symptom-based diagnosis is also vital.[7] The study examines the disease diversity, occurrence, and severity on major crops and its locations to help in proper identification of diseases, advancement of disease control measures and dissemination of sustainable vegetable production technologies.

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Significance of Vegetables in Agricultural Systems

Vegetables are an essential component of sustainable farming as they guarantee nutritional security, assist farm income, and help achieve ecological balance. Yet, limited investment in research and development has contributed to a disparity between what is possible in terms of production with existing varieties, and what is wanted in terms of quality and availability. Neglected indigenous and locally adapted varieties are also nutrient-dense for both humans and animals, and they have been resilient to local conditions. For example, leafy vegetables (i.e., amaranth and cowpea) provide more than the daily recommended

value of vitamin A per 100g of edible portion, which is particularly important for pregnant women. Economically, the area planted in vegetable cultivation is expanding. For instance, tomato production in Jharkhand, India increased from 309.642 tons in 2023 to 311.541 tons in 2024, even though this area has expanded through improved agronomic practices and savings in farm income. The Solanaceae group of crops (e.g., tomato, brinjal and potato) are characterized with heavy incidence of diseases among all varieties shown to the genetic similarity. Tomato Early Blight and potato Late Blight are responsible for severe yield loss.[2] This articles remains to fill gaps of knowledge by conducting field trials and disease characterization in vegetable crops.

MATERIALS & METHODS

Survey and Sample Collection:

A field survey was conducted in four selected areas of the East Singhbhum region of Jharkhand, India: namely Simagoda, Bamni Tola, Pagda, and Parsudih. During the survey, samples of diseased vegetables were collected, including Tomato (*Solanum lycopersicum*), Brinjal (*Solanum melongena*), Cabbage (*Brassica oleracea*), Potato (*Solanum tuberosum*), and Capsicum (*Capsicum annuum*).

RESULT & DISCUSSION

The survey conducted in multiple locations showed the presence of several vegetable crop diseases in tomato, brinjal, cabbage, and potato crops. Of those crops, tomatoes were associated with the greatest number of diseases. In Simagoda, Boram, early blight characterized by dark, concentric lesions and necrotic lesions on leaves were observed in tomatoes. Bamni Tola, Patamda had the highest numbers and complexities of diseases, including leaf curl, anthracnose, and blossom-end rot, resulting in leaf abnormalities and fruit rot. In Pagda, Patamda, brinjal had cucumber downy mildew with powdery patches, cabbage had downy mildew, and late blight caused coalesced brown leaf spots on potatoes. Parsudih recorded ring rot in potatoes characterized by vascular discoloration and concentric lesions in capsicum. In summary, fungal pathogens were recurrent, with varying degrees of symptoms and progressiveness. Symptoms most frequently observed including water-soaked lesions, necrosis and mould proliferation indicated the favourable warm-moist conditions for the manifestation of fungal diseases in these areas.

The results depict substantial pathogens variability of the screened vegetables crops on which Solanaceae, tomato, brinjal and potato are highly susceptible by virtue

Table 1. Symptoms and sign of the Vegetable Crops diseases

Place	Vegetable Sample	Observed Symptoms	Name of Disease	Reference
Simagoda, Boram	**Tomato** (*Solanum lycoperiscum*)	Presence of initial small dark, sunken water-soaked spots on leaves which later become round, brown and have target like concentric rings. Necrosis on leaves.	**Early Blight**	4
Bamni Tola, Patamda	**Tomato** *(Solanum lycoperiscum)*	In-folding of the leaves from the outer side and stunting of the plant. Light Brown color spots on leaves.	**Leaf Curl**	3
		Presence of sunken, dark brown water-soaked lesions in the center.	**Anthracnose**	6
		A dark, sunken, leathery spot on the bottom of the fruit. In time, lesions often become covered with a secondary black mold.	**Blossom-end rot**	8
Pagda, Patamda	**Brinjal** (*Solanum melongena*)	Yellowing of the leaves from the underside of the plants. Presence of chlorotic lesions irregular in shape but later becomes brown in color. Presence of small, brownish angular leaf spot showing infoldings of the leaves from margin.	**Cercospora Leaf Spot**	7
	Cabbage (*Brassica oleracea*)	Presence of small, powdery patches on above ground surfaces first observed on underside of the leaves and later it becomes papery and dies off.	**Downy Mildew**	1
	Potato (*Solanum tuberosum*)	Presence of Sunken, water-soaked spots appearing on leaves which and cover large surface area. later on, becomes brown and coalesced	**Late Blight**	2
Parsudih	**Potato** (*Solanum tuberosum*)	Appearance of sunken cracked patches on the surface of the skin of Potato. Appearance of brown necrosis in the vascular ring.	**Ring Rot**	5
	Capsicum (*Capsicum annuum*)	Presence of concentric circles with black color lesions in the center.	**Anthracnose**	9

with some genetic closeness in relation to their susceptibility factors.[4] It could also be attributed high pathogen pressure or presence of conducive environment. When hydrophytes appear and the blight is general, humidity is very great. Productivity and profitability is compromised by yield-limiting diseases such as Early Blight, Late Blight and Blossom-end rot.[8] The appearance of Cercospora Leaf Spot and Ring Rot suggest an integrated management approach is warranted. Comorbidities demand crop- and site-specific approaches. Symptomatic differentiation at field levels (concentric rings in Early Blight, etc.) is helpful for the precise and effective diagnosis of disease.[7]

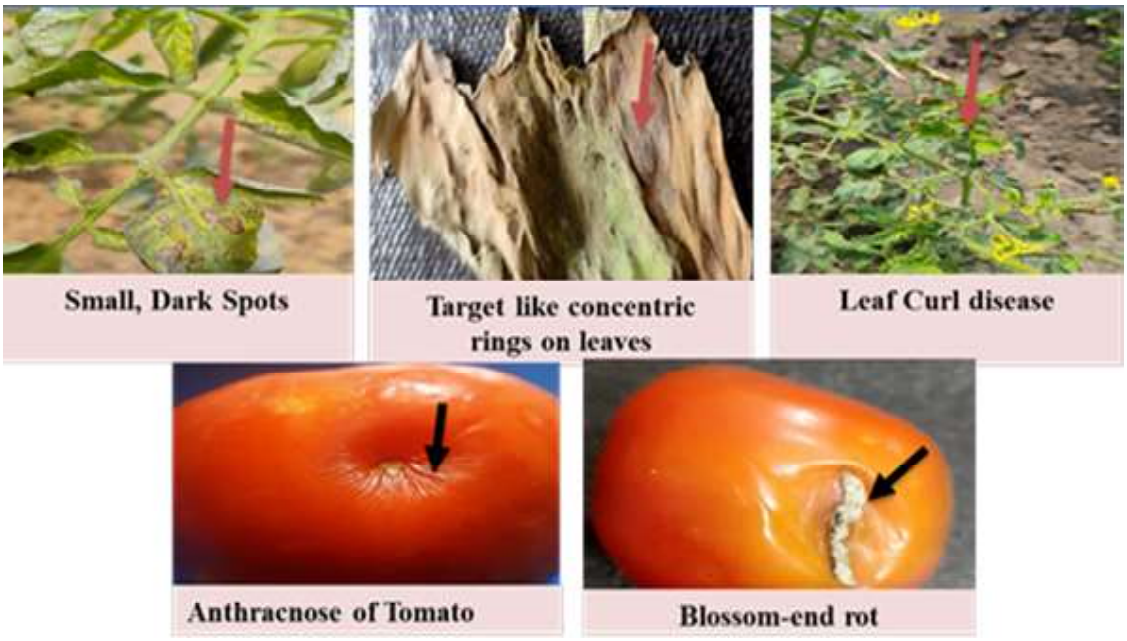


Fig. 1- Symptoms of Disease on Tomato

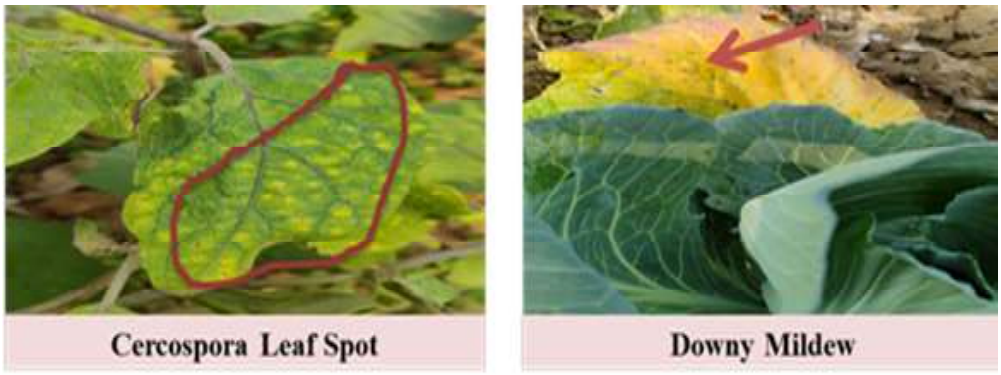


Fig. 2- Symptoms of Disease on Cabbage



Fig. 3- Symptoms of Disease on Potato and *Capsicum*

CONCLUSION

The field survey identified eight primary diseases of vegetable crops across four sites, primarily affecting tomatoes, brinjal, cabbage, and potato. A high proportion of the diseases were caused by fungi, notably in the Solanaceae crops, with tomatoes experiencing high levels of susceptibility. The high occurrence of Early Blight, Late Blight, and Blossom-end rot disease indicates the need for an integrated approach to disease management. The conducive environmental conditions provided ample opportunity for fungal growth, which supports the use of preventive and cultural control strategies. From the baseline study information, site-specific contents, Interventions and knowledge can be tailored to assist farmers with early detection and proper management of the disease. Experiments replicating the field trials should be conducted to isolate pathogens, characterize their molecular features, screen for resistant genotypes and develop an integrated pest management of disease system. Continuous tracking and surveillance would be key for tracking disease trends as part of advancing a sustainable approach to vegetable production and food security within the region.

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